5 **(a)** A beam of vertically polarised light of wavelength 540 nm is incident normally on a diffraction grating, as shown in Fig. 5.1.

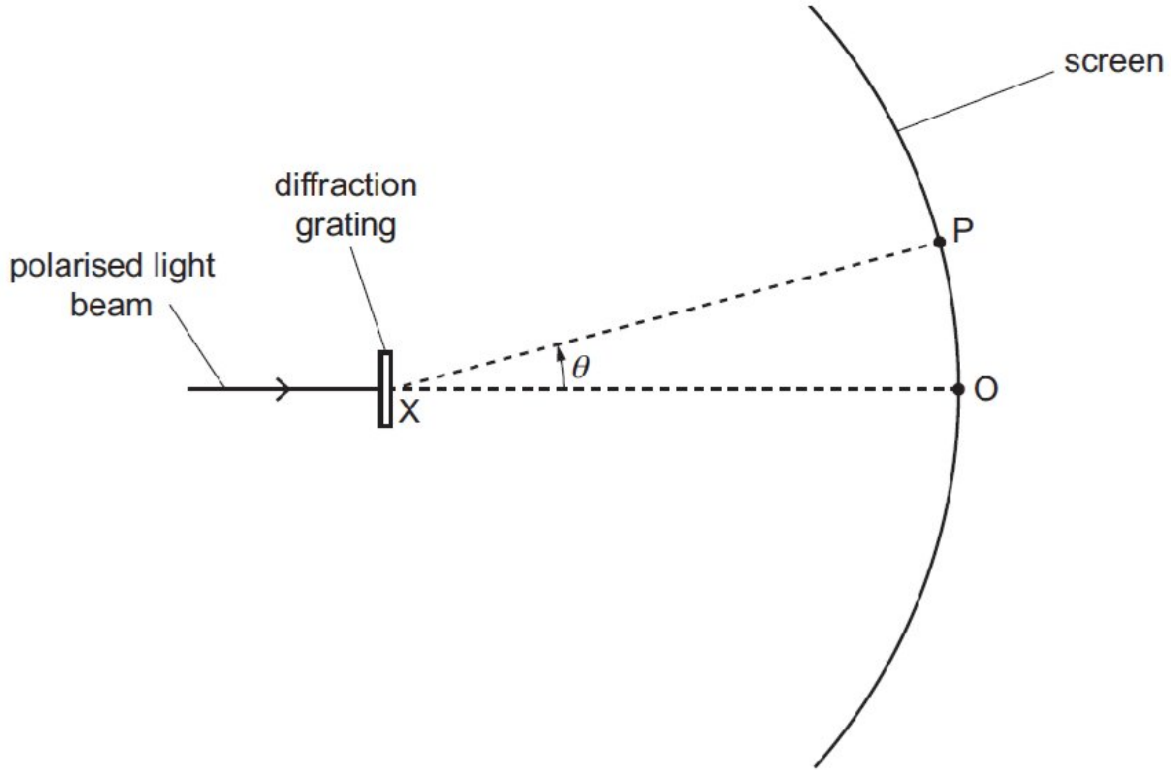


Fig. 5.1 (not to scale)

The diffraction grating has a line spacing of 5.0×10^{-6} m.

The light transmitted by the diffraction grating illuminates a circular screen. The diffraction grating is at the centre X of the circle. The central bright fringe is formed at point O on the screen and has intensity I_0.

P is a point on the screen where the line XP is at a variable angle θ to the line XO. The intensity I of light on the screen at P varies with θ.

(i) Show that the angle θ at which the first-order bright fringe is formed is 6.2°. [1]

(ii) Determine the value of θ at which the second-order bright fringe is formed.

θ = ...° [1]

(iii) On Fig. 5.2, sketch the variation of the intensity I with θ for values of θ from $-15°$ to $+15°$. [3]

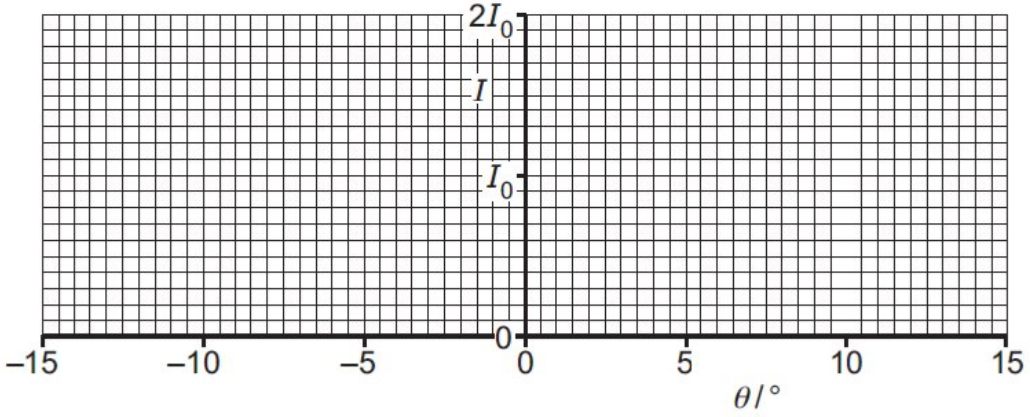


Fig. 5.2

(b) A polarising filter is placed in the path of the light beam that is incident on the diffraction grating in Fig. 5.1. The transmission axis of the filter is at 45° to the vertical.

State how the intensity at the peaks for the light on the screen compares with the answer in **(a)(iii).**

.....

.....

..... [1]

[Total: 6]